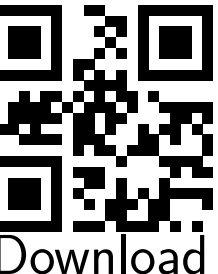




Periodic Orbits of Rotating Kepler Problem: Bifurcation and Conley-Zehnder Indices

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Toy Example : Function



This is graph of $f_t(x) = x(x^2 + t)(x - 1)$ for $t = -1, -1/2, 0$ and 1 .

See the change of critical points.

- $t < 0$, two minima (●) and one maximum. (●)
- $t = 0$, one minimum and one maximum meets at the inflection point. (●)
- $t > 0$, one minimum survives. (●)

This phenomenom also happens for **periodic orbits**.

Bifurcation of Hamiltonian Periodic Orbits

Let $H : W \rightarrow \mathbb{R}$ be a Hamiltonian and $\gamma_c \subset H^{-1}(c)$ be a periodic orbit with energy c .

We call γ_c is **non-degenerate** if the linearized return map on the Poincaré section doesn't have 1 as an eigenvalue.

(Similar to critical point x of a function f such that $f''(x) \neq 0$.)

If γ_c is non-degenerate, there exists a family of periodic orbits near energy c ,

$$\Gamma = \{(\gamma_{c'}, c') : \gamma_{c'} \subset H^{-1}(c'), c - \varepsilon < c' < c + \varepsilon\}.$$

Topologically, Γ is a cylinder, i.e., $\Gamma : S^1 \times (c - \varepsilon, c + \varepsilon) \rightarrow W$.

If γ_c is degenerate, **bifurcation** might happen: γ_c vanishes or another orbits are born.

Generically, there are 8 types of bifurcations of Hamiltonian periodic orbits. ([AM78])

Another kind of bifurcation may happen if there is enough symmetry. (e.g. RKP)

Conley-Zehnder Index and Floer Homology

The **Conley-Zehnder index** $\mu_{CZ}(\gamma)$ is an integer associated to the periodic orbit. ([CZ83])

If the energy hyperplane $H^{-1}(c)$ has a contact structure, the **symplectic homology** $SH_*(W_c)$, generated by periodic orbits, can be defined for each energy c . ([FH94], [CFH95])

The circular restricted three-body problem have such property. ([AFvKP12], [CJK20])

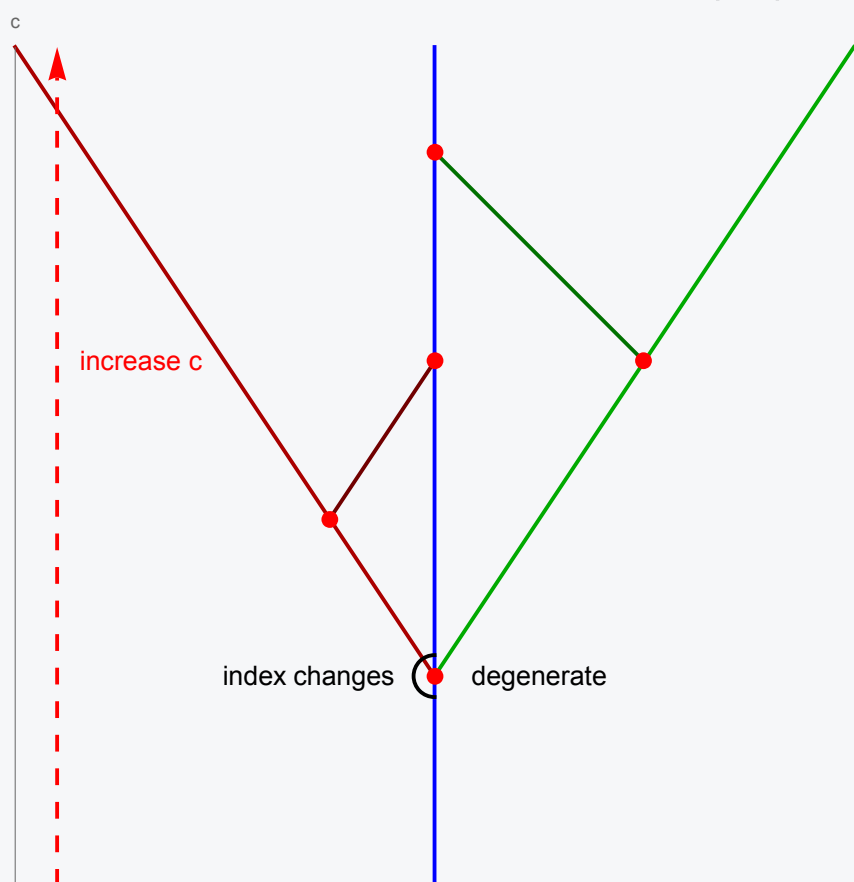
- $SH_*(W_c)$ is invariant of c , until c passes a critical energy.
- The Conley-Zehnder index gives a grading to $SH_*(W_c)$.

It means that the indices of *all* periodic orbits are globally controlled.

Computing tool : Local Floer homology, Morse-Bott spectral sequence.

Index and Bifurcation

$\mu_{CZ}(\gamma_c)$ is invariant while γ_c is non-degenerate, and might change when γ_c degenerates. In this case, the change of $\mu_{CZ}(\gamma_c)$ is related to the index of other orbits.



Toy model of a bifurcation

Periodic orbits with same energy c has the same y -coordinates.

There's only one orbit ● for low c .

When ● passes degeneracy (●), new orbits (●, ●) are born.

$SH(W)$ generated by ● at low energy and ●, ●, ● at higher energy must coincide, so the Conley-Zehnder indices of these orbits are related.

This happens until c passes critical energy.

Periodic Orbits of Kepler Problem

Kepler problem is defined by Hamiltonian $E : T^*\mathbb{R}^3 \setminus \{0\} \rightarrow \mathbb{R}$,

$$E(q, p) = \frac{|p|^2}{2} - \frac{1}{|q|}.$$

If $E < 0$, every orbit is an ellipse with one focus at the origin.

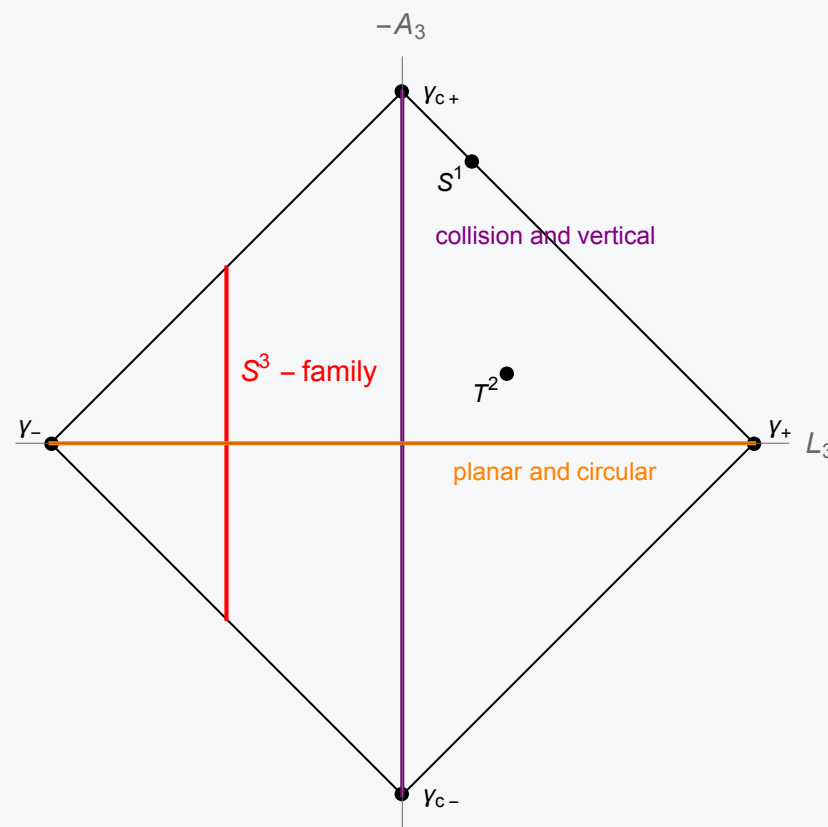
We resolve the singularity at 0 by include collision orbits via **Moser regularization**.

The orbits have two invariants: angular momentum L and **Laplace-Runge-Lenz vector**

$$A = p \times L - \frac{q}{|q|}.$$

The space of periodic orbits with Kepler energy E can be parametrized by a map

$$\begin{aligned} \Phi : \mathcal{M}_E &\rightarrow S^2 \times S^2 \\ \gamma &\mapsto (\sqrt{-2E}L - A, \sqrt{-2E} + A) \end{aligned}$$



The diagram is an image of the map

$$\begin{aligned} \mathcal{M}_E &= S^2 \times S^2 \rightarrow \mathbb{R}^2 \\ \gamma &= (x, y) \rightarrow \frac{1}{2}(x_3 + y_3, x_3 - y_3). \end{aligned}$$

Four vertices correspond to the retrograde, direct and vertical collision orbits.

The family of orbits with same L_3 forms S^3 -family.

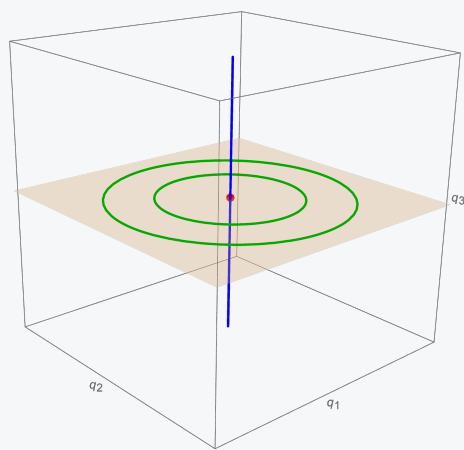
The space is same as the space of unit geodesics of round S^3 .

Periodic Orbits of Rotating Kepler Problem

The **rotating Kepler problem** is defined by Hamiltonian

$$H(q, p) = \frac{|p|^2}{2} - \frac{1}{|q|} + (p_1 q_2 - p_2 q_1) = E + L_3.$$

This is a limit $\mu = 0$ of the circular restricted three-body problem (CRTBP).

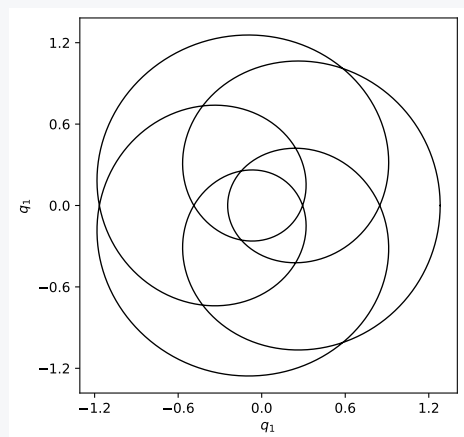


Since $\{E, L_3\} = 0$, the orbits which is invariant under the rotation along q_3 -axis are always periodic.

Planar circular (○): **retrograde** (γ_+) and **direct** (γ_-).

Vertical collision (—): northern (γ_{c+}) and southern (γ_{c-}).

These are non-degenerate for generic energy $c < -1.5$.



For $E_{k,l} = -0.5(k/l)^{2/3}$ for $k, l = 1, 2, \dots$, the Kepler flow and rotation resonate and every orbit becomes periodic.

These orbits form S^3 -family $\Sigma_{k,l}$ for each $E_{k,l}$.

At these energies, $\gamma_{\pm}$ and $\gamma_{c\pm}$ become degenerate.

$\Sigma_{k,l}$ is born at γ_-^{k-l} and vanishes at γ_+^{k+l} .

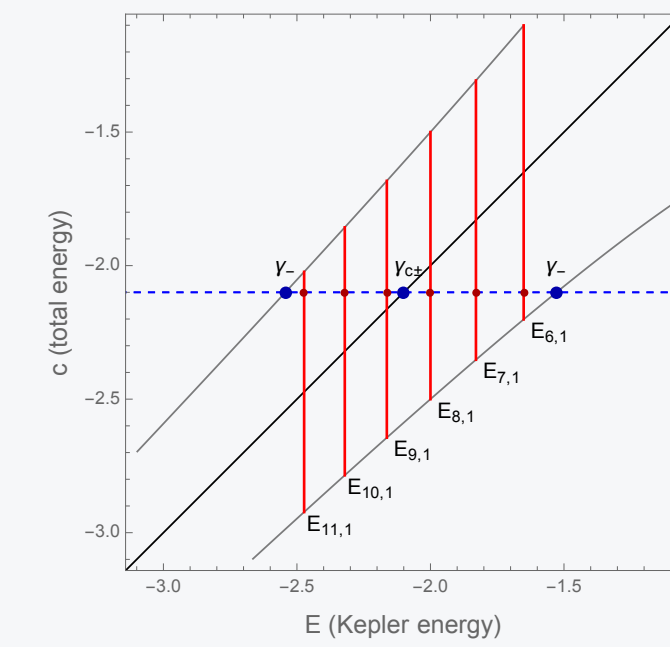
- The simple retrograde orbit doesn't bifurcate if $c < -1.5$.
- N -th cover of retrograde orbit bifurcate $(N - 1)$ -times.
- Every cover of direct orbit bifurcate infinitely many times.
- Vertical collision orbits don't bifurcate, due to too much symmetry.

[AFFvK13] analyzed the planar case.

This is also observed in CRTBP, but there are much less symmetry.

$\Sigma_{k,l}$ become a finite set of discrete orbits, and the vertical collision also bifurcates.

Conley-Zehnder Indices of Rotating Kepler Problem



At generic energy c , the energy hyperplane $H^{-1}(c)$ ($\dots$) contains following orbits:

- 4 non-degenerate orbits: $\gamma_{\pm}$ and $\gamma_{c\pm}$.
- $\Sigma_{k,l}$ -family for each $E_{k,l}$ such that

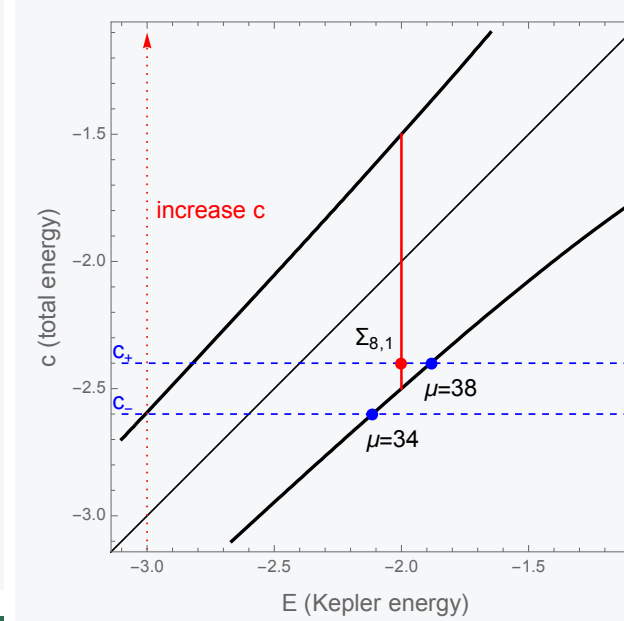
$$E_{k,l} - (-2E_{k,l})^{-1/2} < c < E_{k,l} + (-2E_{k,l})^{-1/2}.$$

- These are all generators of $SH^{S^1,+}(T^*S^3)$, and their indices can be investigated with local Floer homology.
- $SH^{S^1,+}(T^*S^3)$ has one generator at degree 2 and two generators at each degree $2k \geq 4$.

Theorem ([Lee25], ArXiv Preprint)

Let $c_{k,l}^{\pm} = E_{k,l} \pm (-2E_{k,l})^{-1/2}$, and denote N -th cover of orbit γ by γ^N .

- $\mu_{CZ}(\gamma_+^N) = 4N - 2$ for $c < c_{N-1,1}^+$, and decreases by 4 each time c passes $c_{N-k,k}^+$.
- $\mu_{CZ}(\gamma_-^N) = 4N + 2$ for $c < c_{N+1,1}^-$, and increases by 4 each time c passes $c_{N+k,k}^-$.
- $\mu_{CZ}(\gamma_{c\pm}^N) = 4N$ for any $c < -1.5$.
- $\mu_{RS}(\Sigma_{k,l}) = 4k - 1/2$.



This behavior can be analyzed via local Floer homology and Morse-Bott spectral sequence.

c passes $c_{k,l}^-$: $\mu_{CZ}(\gamma_-^{k+l})$ increases by 4, $\Sigma_{k,l}$ is born.

c passes $c_{k,l}^+$: $\mu_{CZ}(\gamma_+^{k-l})$ decreases by 4, $\Sigma_{k,l}$ vanishes.

The change of $\mu_{CZ}(\gamma_{\pm}^N)$ must change $SH_*(W)$, but $\Sigma_{k,l}$ corrects the difference.

Discussion : Finding Periodic Orbits

[AFvK⁺] analyzed Hill's lunar problem in this framework with numerical computation.

Goal: Apply this framework to analyze the periodic orbits of circular restricted three-body problem, and other interesting systems.

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